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Date: 18/03/2026

Duration: 1 hour 45

CONTINUOUS ASSESSMENT 1

II YEAR – SEMESTER 4

CSE23AE204 - PCP III

(B.Tech. E01,E02,E03,E05,E06, E61)

SET A

Question 1: Problem Statement

A company provides charging facilities for multiple electronic devices throughout the day.

Each device requires charging for a specific **time interval**, represented as:

[start, end]

where:

- **start** → time when charging begins
- **end** → time when charging finishes

All charging sessions are given as a list of intervals:

intervals = [[start1, end1], [start2, end2], ...]

A **charging port can charge only one device at a time**. If two charging sessions overlap in time, they must use **different ports**.

Your task is to determine the **minimum number of charging ports required** so that all devices can be charged **without any waiting time**.

Example 1

Input

intervals = [[1,4],[2,5],[7,9]]

Output

2

Explanation

Device charging schedule:

Device1 : [1-----4]

Device2 : [2-----5]

Device3 : [7---9]

Between time **2 and 4**, two devices require charging simultaneously.

Therefore, **2 charging ports** are required.

Question 2: Problem Statement

You are given a JSON object:

```
{
  "jobs": [[id, deadline, profit], ...]
}
```

Each job has:

- **id** → Unique job identifier
- **deadline** → The last time slot by which the job must be completed
- **profit** → Profit earned if the job is completed

Example 1:

```
{
  "jobs": [
    [1, 4, 20],
    [2, 1, 10],
    [3, 1, 40],
    [4, 1, 30]
  ]
}
```

Output

[2, 60]

Rules:

- Each job takes exactly **1 unit of time**.
- Only **one job can be executed at a time**.
- A job must be completed **on or before its deadline**.

Your task is to schedule jobs to maximize total profit.

Output Format

Return:

[count, profit]

Where:

- count → Number of jobs successfully scheduled
- profit → Maximum total profit earned

Explanation

Step 1: Sort jobs in descending order of profit:

[3,1,40]

[4,1,30]

[1,4,20]

[2,1,10]

Step 2: Schedule jobs in latest possible free slot:

- Job 3 → Slot 1
- Job 4 → Cannot schedule (slot 1 already filled)
- Job 1 → Slot 4
- Job 2 → Cannot schedule

Total scheduled jobs = 2

Total profit = 60

Question 3: Problem Statement**Problem Statement**

You are given a list of ropes where each rope has a certain **length**.

You want to connect all ropes into **one single rope**.

When you connect two ropes of lengths a and b:

cost = a + b

After connecting them:

new rope length = a + b

This newly formed rope can again be used to connect with other ropes.

Your goal is to determine the **minimum total cost required to connect all ropes into one rope**.

Key idea:

To minimize cost, you should **always connect the two smallest ropes first**.

Example 1

Input

ropes = [4,3,2,6]

Output

29

Explanation

Step-by-step:

2 + 3 = 5 cost = 5

ropes → [4,5,6]

4 + 5 = 9 cost = 9

ropes → [6,9]

6 + 9 = 15 cost = 15

Total cost:

5 + 9 + 15 = 29